

Laws of Indices — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong. Almost every note below comes back to one of three errors: multiplying the base as well as combining the indices, applying a law where the bases do not match, or reading a negative index as a negative answer. Watch which of the three a given student favours — they rarely make all of them, and the one they make is the one to drill.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** Row 1 $2^5 \rightarrow 2 \times 2 \times 2 \times 2 \times 2$ and **32** [6]
 Row 2 $3^3 \rightarrow 3 \times 3 \times 3$ and **27**
 Row 3 $10^4 \rightarrow 10 \times 10 \times 10 \times 10$ and **10 000**
One mark per correct box. Watch for the base multiplied by the index in the value column: $2 \times 5 = 10$, $3 \times 3 = 9$ (right for the wrong reason) and $10 \times 4 = 40$. A student who writes 10 for 2^5 has the misconception the whole topic is built to remove.
- 2** (a) 3^5 [2]
 (b) $2^3 \times 7^2$
In (b) the two bases are different, so they stay separate. Any attempt to combine them into a single power scores zero — there is nothing to count once the factors stop being identical.
- 3** (a) $5 \times 5 \times 5 = 125$ [2]
 (b) **128**
The pair is deliberate: 125 and 128 sit next to each other, so a student who has memorised “ 2^7 is about 125” rather than counted it will not survive the comparison.
- 4** (a) $4 + 3 = 7$, so 2^7 [2]
 (b) $2 + 5 = 7$, so 7^7
Answers of 2^{12} and 7^{10} mean the indices were multiplied. That is the next law, not this one.
- 5** (a) $8 - 5 = 3$, so 6^3 [2]
 (b) $7 - 4 = 3$, so 10^3
- 6** (a) a^8 [2]
 (b) b^4
- 7** (a) c^9 [3]
 (b) d^6
 (c) e is e^1 , so $3 + 1 + 2 = 6$, giving e^6
Part (c) is the whole question. A letter written with no index has an index of 1, and students who read it as 0 give e^5 . Ask what e on its own is worth before correcting the arithmetic.
- 8** The bases are different, so there is nothing to count — the laws only apply when the base is the same [1]
 $8 \times 25 = 200$ [1]
Accept any wording that names the mismatched bases. An answer of 10^5 (bases multiplied, indices added) is the failure this question exists to catch — it gives 100 000 instead of 200.
- 9** $3^4 = 81$ and $4^3 = 64$ [1]
 3^4 is larger [1] [2]
Both values must be shown for the first mark. The point is that swapping the base and the index is not a harmless move — worth saying out loud, because 2^4 and 4^2 are equal and students generalise from that one accident.

- 10** (a) $2^3 \times 2^4 = 2^7$ [1]
 (b) $2^7 \div 2^5 = 2^2$ [1]
 (c) **4** [1]
The long way is $8 \times 16 = 128$, then $128 \div 32 = 4$ — same answer, three-figure numbers. Award full marks for it, but point out what the laws saved.
- 11** The base was multiplied as well as the indices being added; only the indices are added, [2]
 and the base stays as it is [1]
 Correct answer: $2^8 (= 256)$ [1]
 4^8 is 65 536 against a true answer of 256 — a factor of 256 out. A student who checks the size of an answer catches this without knowing any law.
- 12** $64 = 2^6$ [1]
 $64 = 4^3$ [1]
Both are correct and both are wanted. This is the first hint of Paper C: a number can be rewritten in whichever base makes the question work.

Paper B — Medium

30 MARKS • NON-CALCULATOR

- 1** (a) $3 \times 2 = 6$, so 2^6 [3]
 (b) 5^{12}
 (c) x^{10}
 2^5 , 5^7 and x^7 are the added-instead-of-multiplied answers. The fix is to read the outer index as “how many copies of the bracket”.
- 2** (a) $3^2 = 9$ and $(x^2)^2 = x^4$, so $9x^4$ [3]
 (b) $2^4 = 16$ and $(y^3)^4 = y^{12}$, so $16y^{12}$
 (c) $125a^3$
 $3x^4$, $2y^{12}$ and $5a^3$ are the standard errors: the law was applied to the letter and the coefficient was left sitting there. The bracket does not care which parts are numbers.
- 3** (a) $a^{12} \div a^5 = a^7$ [2]
 (b) $b^6 \times b^4 = b^{10}$
Two laws in one line. Marking is all-or-nothing per part, but note which of the two steps failed — it is almost always the bracket.
- 4** (a) $2^6 = 64$ [2]
 (b) $10^6 = 1\,000\,000$
- 5** $2^5 = 32$ [4]
 $5^3 = 125$
 $7^0 = 1$
 $10^4 = 10\,000$
One mark per box. $7^0 = 0$ is the answer to watch for, and it is worth stopping on: it is not a slip, it is a belief, and question 6 is the argument that dislodges it.
- 6** $5^n \div 5^n = 5^{n-n} = 5^0$ by the division law (any example such as $5^4 \div 5^4$ is fine) [1]
 But any number divided by itself is 1, so $5^0 = 1$ [1]
“Because that is the rule” scores nothing. The mark is for the two statements being about the same calculation — that is what makes the answer forced rather than arbitrary.
- 7** (a) $1 + 1 = 2$ [3]
 (b) $1 \times 4^3 = 64$
 (c) **1**
(a) catches the student who thinks $a^0 = 0$ and answers 0. (c) catches the one who reads the outer zero as belonging to the 3 — anything to the power zero is 1, however complicated the base looks.

- 8** (a) $2^{4+3-5} = 2^2 = 4$ [3]
 (b) $3^{5-3} = 3^2 = 9$
 (c) $10^6 \div 10^4 = 10^2 = 100$
Full marks for a correct value however it was reached, but a student who evaluates $3^5 = 243$ and then divides has not understood what the laws are for. Say so.
- 9** (a) $3 \times 4 = 12$ and a^{2+5} , so **$12a^7$** [3]
 (b) $(2p^3)^2 = 4p^6$, times p gives **$4p^7$**
 (c) $20 \div 5 = 4$ and b^{8-2} , so **$4b^6$**
Coefficients multiply and divide as ordinary numbers while the indices add and subtract. Students who add the coefficients in (a) give $7a^7$; that is the tell that the two halves of the term are being treated the same way.
- 10** (a) **64** [3]
 (b) **32**
 (c) In (a) the outer index says two copies of 2^3 , so the indices multiply to give 2^6 ; in (b) two powers are joined, so the indices add to give 2^5 **[1]**
The two answers differ by a factor of 2, which is small enough that a student who muddles the laws will not notice from the size alone. The explanation in (c) is the mark that matters.
- 11** $2^{10} = 1024$ and $10^3 = 1000$ **[1]** [2]
 2^{10} is larger [1]
They are within 24 of each other, so guessing is no help — the working has to happen. Doubling ten times is also worth practising in its own right; $2^{10} = 1024$ turns up everywhere later.

Paper C — Hard

30 MARKS • NON-CALCULATOR

- 1** (a) $1/2^4 = 1/16$ [3]
 (b) $1/5^2 = 1/25$
 (c) $1/10^3 = 1/1000$
–16, –25 and –1000 are the answers to watch for. The minus sign is an instruction to flip the power over, not a sign attached to the answer. Every value here is positive and less than 1.
- 2** (a) $9 = 3^2$, so $1/9 = 3^{-2}$ [3]
 (b) $32 = 2^5$, so $1/32 = 2^{-5}$
 (c) $100 = 10^2$, so $1/100 = 10^{-2}$
This is question 1 run backwards, which is the harder direction. A student who can do 1 but not 2 has learned a procedure rather than the meaning.
- 3** (a) $3 - 7 = -4$, so 2^{-4} **[1]**, which is **$1/16$ [1]** [3]
 (b) $8/128$, and cancelling gives **$1/16$** — the two routes agree **[1]**
Part (b) is the proof that the negative index is not a convention someone invented. Both routes have to agree, and seeing them agree once is what makes the rule usable.
- 4** (a) **2^2** [3]
 (b) **2^3**
 (c) $4^2 = (2^2)^2 = 2^4$, so $2^5 \times 2^4 = 2^9$
Check: $32 \times 16 = 512 = 2^9$. Writing 8^7 in (c) — bases multiplied, indices added — is the usual wrong route. Parts (a) and (b) are there to make the conversion obvious; a student who does them and then ignores them has not seen why they were asked.

- 5 (a) $9 = 3^2$, so $9^3 = (3^2)^3 = 3^6$ [1] [4]
 (b) $3^6 \times 3^2 = 3^8$ [1]
 (c) $27 = 3^3$ and $9^2 = 3^4$, so $3^{3+4-5} = 3^2$ [1]
 $= 9$ [1]

Every term must be converted to base 3 before any law is used. In (c), a student who converts 27 but forgets 9^2 gets $3^3 \times 9^2 \div 3^5$ and stalls — award the conversion mark and point at the term still in the wrong base.

- 6 (a) $(2a^3)^2 = 4a^6$ [1], times $3a^4$ gives $12a^{10}$ [1] [4]
 (b) $(2x^2)^3 = 8x^6$ [1], so $24x^7 \div 8x^6 = 3x$ [1]

The first mark in each part is the bracket, and it is where both go wrong: $2a^6$ in (a) and $2x^6$ or $6x^6$ in (b). Award the bracket mark separately so the pattern shows. In (b), $3x^1$ is written $3x$ — do not penalise either form.

- 7 $16 = 2^4$, so $16^2 = 2^8$ [1] [3]
 $2^8 \div 2^5 = 2^3$ [1]
 $= 8$ [1]

Check the long way: $256 \div 32 = 8$. Doing it that way is correct and scores full marks, but it is the version that stops working the moment the numbers grow.

- 8 (a) $-3 + 5 = 2$, so $2^2 = 4$ [1] [1] [4]
 (b) $4 + (-6) = -2$, so $3^{-2} = 1/9$ [1] [1]

One mark for the combined index, one for the value. The laws behave identically with negative indices — the arithmetic on the indices is just ordinary directed number. A student who is fine here but not on question 1 has an index problem, not a negative-number problem.

- 9 $2^{-1} = 1/2$, $3^0 = 1$, $2^{-3} = 1/8$, $3^1 = 3$ [1] [1] [3]
 Order: 2^{-3} , 2^{-1} , 3^0 , 3^1 [1]

Two marks for the four values (one if two or three are right), one for the order. A student who ranks by the index alone puts 2^{-3} and 2^{-1} the wrong way round only if they also forget the flip; the flip reverses the order of the negatives, which is the point of including two.

Paper D — Hard

30 MARKS · NON-CALCULATOR

- 1 $1000 = 10^3$ [4]
 $1\ 000\ 000 = 10^6$
 $0.01 = 1/100 = 10^{-2}$
 $0.0001 = 1/10\ 000 = 10^{-4}$

One mark per box. The two decimals are where it goes wrong, and almost always by one: 0.01 read as 10^{-1} because it has one zero after the point. Count the places, or write the fraction down first.

- 2 (a) 10^7 [3]
 (b) $5 - 8 = -3$, so 10^{-3}
 (c) **0.001**

In (b) the top index is smaller than the bottom one, which is exactly the case students avoid by silently swapping them to get 10^3 . The decimal in (c) catches it: 1000 is not a small number.

- 3 (a) $10^{-4} \times 10^3 = 10^{-4+3}$ [1] = 10^{-1} m [1] [4]
 (b) **0.1 m**, which is **10 cm** [1]
 (c) The indices were multiplied instead of added — this is a multiplication of two powers, so the law is add [1]

10^{-12} m is about a hundredth of an atom, for a stack of a thousand sheets of paper. The sanity check is free and nobody does it: a thousand sheets is roughly a ream, which is about 10 cm high.

- 4** (a) $6^4 \div 6^4 = 6^{4-4} = 6^0$ by the division law **[1]** [4]
 But $1296 \div 1296 = 1$, and both describe the same calculation, so $6^0 = 1$ **[1]**
 (b) Nothing in the argument used the 6 — $a^n \div a^n$ is 1 for any base a **[1]**
 (c) With base 0 the calculation is $0 \div 0$, which has no value, so the argument gives nothing — this is why 0^0 is excluded **[1]**
Part (c) separates the student who has followed the argument from the one who has memorised its conclusion. Accept any answer identifying the division by zero. Do not require the phrase “undefined”.
- 5** (a) $2 - 5 = -3$, so 5^{-3} **[1]** [4]
 (b) $25/3125$ **[1]**, and cancelling 25 from both gives $1/125$ **[1]**
 (c) The two answers describe the same calculation, so $5^{-3} = 1/5^3$ — a negative index means the **reciprocal** **[1]**
This is the negative-index rule being derived rather than quoted, and it is the same shape of argument as question 4. A student who can produce both has understood that the two odd-looking rules are consequences of the division law, not extra rules bolted on.
- 6** $10^{-2} = 1/100$, $2^{-3} = 1/8$, $5^{-1} = 1/5$, $3^{-2} = 1/9$ **[1] [1]** [3]
 Order: 10^{-2} , 3^{-2} , 2^{-3} , 5^{-1} **[1]**
 $1/9$ and $1/8$ are close (0.111 and 0.125) and are the pair that gets swapped. Comparing denominators is enough — the bigger the bottom, the smaller the fraction — and no decimals are needed. A student who orders by the index gives 5^{-1} first, which is the largest.
- 7** (a) $(3x^2)^3 = 27x^6$ **[1]**, so $27x^6 \div 9x^4 = 3x^2$ **[1]** [4]
 (b) Inside first: $a^5 \times a^{-2} = a^3$ **[1]**, then $(a^3)^2 = a^6$ **[1]**
In (a) the cube of the 3 is the mark that goes missing — $3x^6 \div 9x^4$ gives a fraction and the student usually assumes they have gone wrong somewhere else. In (b), $5 + (-2) = 3$ is directed number, not an index rule.
- 8** (i) The mistake: the indices were added. A power of a power multiplies them **[1]** [4]
 Correct answer: x^6 **[1]**
 (ii) The mistake: a negative index means the reciprocal, not a negative answer **[1]**
 Correct answer: $1/10^2 = 1/100$, or 0.01 **[1]**
These are the two errors the topic is built around, which is why they close the paper. Marking someone else's working is identical cognitive work to checking your own and far easier to face — a student who can diagnose both here will hesitate before writing either.